

Ex. An electron has a speed of 300 m/s accurate to 0.01%. With what fundamental accuracy can we locate the position of the electron?

Solution. The momentum of the electron

$$p = mv = 9.1 \times 10^{-31} \times 300 = 27.2 \times 10^{-29} \text{ kg-m/s.}$$

The uncertainty Δp in the momentum is given to be 0.01% of this value,

$$\begin{aligned} \therefore \Delta p &= \frac{0.01}{100} \times 27.3 \times 10^{-29} \\ &= 27.3 \times 10^{-33} \text{ kg-m/s.} \end{aligned}$$

Therefore, the minimum uncertainty in position of electron

$$\Delta x = \frac{h}{\Delta p} = \frac{6.62 \times 10^{-34}}{27.3 \times 10^{-33}} = 2.42 \times 10^{-2} \text{ m.}$$

Ex. 1 Calculate the uncertainty in the velocity of an electron which is confined in a 1 Å box.

Solution. The uncertainty in the momentum of the electron

$$\begin{aligned} \Delta p &= \frac{h/4\pi}{\Delta x} \\ &= \frac{6.62 \times 10^{-34} \text{ Joule sec.} / (4 \times 3.14)}{1 \times 10^{-10} \text{ m}} \\ &= 0.52 \times 10^{-24} \text{ kg-m/s} \end{aligned}$$

$\therefore$ Uncertainty in the velocity

$$\begin{aligned} &= \frac{\text{uncertainty in momentum}}{\text{mass of the electron}} \\ &= \frac{0.52 \times 10^{-24}}{9.1 \times 10^{-31}} = 5.8 \times 10^4 \text{ m/s} \end{aligned}$$

Ex. 2 Calculate the uncertainty in the position of an electron which has been accelerated through a potential difference of $V = (1000 \pm 6)$ volts.

Soln. The momentum of the particle

$$p = \frac{h}{\lambda} = \frac{h}{h/\sqrt{(2meV)}} = (2meV)^{1/2}$$

$$\therefore \Delta p = \frac{me \cdot \Delta V}{(2meV)^{1/2}}$$

Also from uncertainty principle

$$\Delta p \cdot \Delta x \approx h$$

$$\text{or } \Delta x = \frac{h}{\Delta p} = \frac{h}{\frac{me \cdot \Delta V}{(2meV)^{1/2}}} = \frac{h}{\left(\frac{me}{2}\right)^{1/2}} \frac{V^{1/2}}{\Delta V}$$

$$\text{or } \Delta x = \frac{6.62 \times 10^{-34}}{\left(\frac{9.1 \times 10^{-31} \times 1.6 \times 10^{-19}}{2}\right)^{1/2}} \times \frac{V^{1/2}}{\Delta V}$$

$$= 0.245 \frac{V^{1/2}}{\Delta V} \text{ \AA.}$$

Since the maximum uncertainty in the accelerating voltage is $2 \times 6.0 = 12.0$ volts.

$$\therefore \Delta x = \frac{0.245 \times (1000)^{1/2}}{12} = 0.65 \text{ \AA.}$$

Ex. 22 The width of a spectral line of $\lambda = 5000 \text{ \AA}$ is found to be $10^{-4} \text{ \AA}$. Compute the Minimum time required for the atomic system to retain the corresponding excitation energy.

Soln. From Uncertainty principle

$$\Delta E \cdot \Delta t = h$$

$$\text{or } \Delta t = \frac{h}{\Delta E}$$

$$\text{since } E = h\nu = \frac{hc}{\lambda}$$

$$\Delta E = -\frac{hc}{\lambda^2} d\lambda$$

$$\therefore \Delta t = \frac{h}{\frac{hc}{\lambda^2} d\lambda} = \frac{\lambda^2}{c \cdot d\lambda}$$

Here, the width of spectral line

$$d\lambda = 10^{-4} \text{ \AA} = 10^{-14} \text{ m}$$

$$\text{and } \lambda = 5000 \text{ \AA} = 5 \times 10^{-7} \text{ m}$$

$$\therefore \Delta t = \frac{(5 \times 10^{-7})^2}{3 \times 10^8 \times 10^{-14}} = \frac{25 \times 10^{-14}}{3 \times 10^{-6}}$$

$$= 8.3 \times 10^{-8} \text{ sec.}$$

Ex 23. An excited atom has a average life time of 10^{-8} sec, that is, during this period, it emits a photon and returns to the ground state. What is the minimum uncertainty in the frequency of this photon ?

Soln. The uncertainty relation for the energy and time is

$$\Delta E \cdot \Delta t \geq \frac{h}{4\pi}$$

Since photon energy $E = h\nu$

Hence above relation becomes

$$\Delta v \cdot \Delta t \geq \frac{1}{4\pi}$$

or

$$\Delta v \cdot \Delta t \geq \frac{1}{4\pi \cdot \Delta t}$$

Here

$$\Delta t = 10^{-8} \text{ sec.}$$

$\therefore$

$$\Delta v \geq \frac{1}{4 \times 3.14 \times 10^{-8}} \geq 8 \times 10^6 \text{ sec}^{-1}$$

Therefore, **minimum uncertainty in frequency is $8 \times 10^6 \text{ sec}^{-1}$.**

4 DUAL NATURE OF LIGHT AND MATTER

The phenomenon of reflection, refraction, interference, diffraction and polarisation of light prove that the light possesses '**wave nature**'. On the other hand photo-electric effect, Compton effect and the emission absorption of radiation are the phenomena which prove that light possesses quantum or **particle nature**. The quanta of light, which are photons according to quantum theory, are shown to possess wave. The diffraction effects exhibited by photons were shown by Taylor's experiments. Whole of this situation led to consider light to possess **dual nature**, *i.e.*, it behaves like a wave and a particle at the same instant.

The quality of light also led to think that the matter too possesses dual nature, *i.e.*, a matter particle behaves as a particle and a wave at the same moment. At atomic levels the determination of e/m in case of electrons, protons and ionised atoms provide the experimental tests about the particle or quantum nature. The working of cathode ray tubes, the electron valves are the day-today examples to illustrate the particle nature of electrons. On the other hand the diffraction experiments of Davisson and Germer, G. P. Thomson, straight edge diffraction, double slit interference, prove that electrons, protons, neutrons and other atomic particles possess wave nature. The most important example is that of electron microscope, which is a practical reality today and works on the basis of the wave nature associated with electrons. This whole situation led to think that the matter also possesses dual nature. The result of duality of light and matter gave rise to wave-mechanics. This mechanics is widely applicable at atomic and nuclear dimensions. It was independently formulated by Schrodinger and Heisenberg under the names of wave-mechanics and quantum mechanics.